Comprehensive E-Waste Assessment Report

State E-Waste Management Tracker | Detailed Analysis

54.2	Needs Improvement	4,816,690 kg
TOTAL SCORE (OUT OF 100)	PERFORMANCE LEVEL	TOTAL STATE GENERATION

1. Executive Summary

This comprehensive report provides a deep-dive analysis of the state's e-waste generation, collection performance, and regulatory compliance based on the Q3 assessment cycle. The overall performance score stands at **54.2/100**, indicating that the state's operations currently fall under the **Needs Improvement** category.

While downstream formal processing and recycler capacities are robust, the upstream collection mechanisms and informal sector integration are severely lagging. A significant portion of generated e-waste is not entering the formal authorized collection channels.

2. Scorecard Performance Breakdown

The 100-point tracker evaluates 7 core categories. The chart below illustrates the points earned against the maximum possible weight for each category.

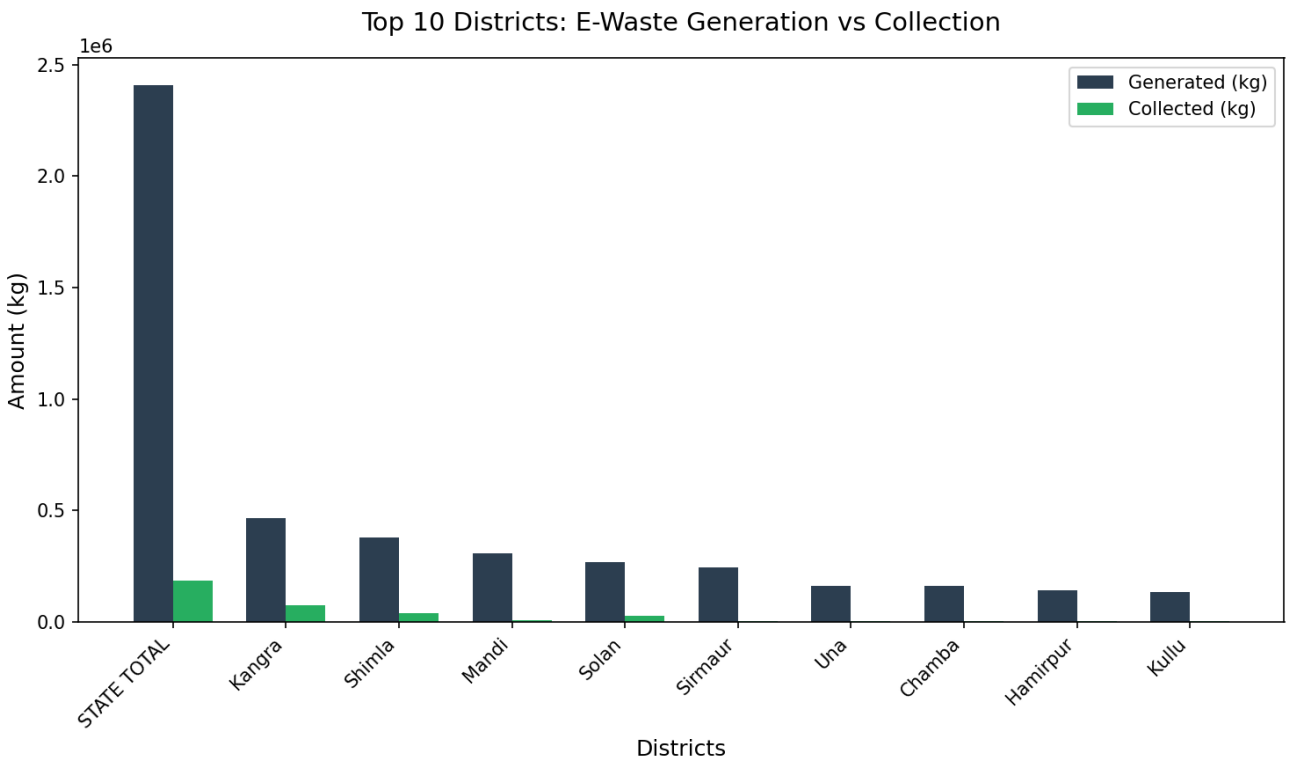
Scorecard Chart

Category	Weight	Score Earned	Status / Observation
Generation & Inventory	15.0	**0.0**	**Critical Gap**
Collection Performance	20.0	**2.9**	**Critical Gap**
Formal Recycling & Treatment	20.0	**20.0**	**Optimal**
Regulatory Compliance	15.0	**11.3**	**Adequate**
Informal Sector Integration	10.0	**0.0**	**Critical Gap**

Category	Weight	Score Earned	Status / Observation
Environmental & Health Safeguards	10.0	**10.0**	**Optimal**
Awareness & Outreach	10.0	**10.0**	**Optimal**
Excellent	nan	**nan**	**Critical Gap**
Good	nan	**nan**	**Critical Gap**
Needs Improvement	nan	**nan**	**Critical Gap**
Critical	nan	**nan**	**Critical Gap**

3. District-Wise Generation and Collection Analysis

Effective e-waste management requires localized infrastructure. The following data highlights the disparity between the e-waste generated at the district level and the volumes successfully captured by registered collection centers and EPR entities.



Detailed District Inventory Data

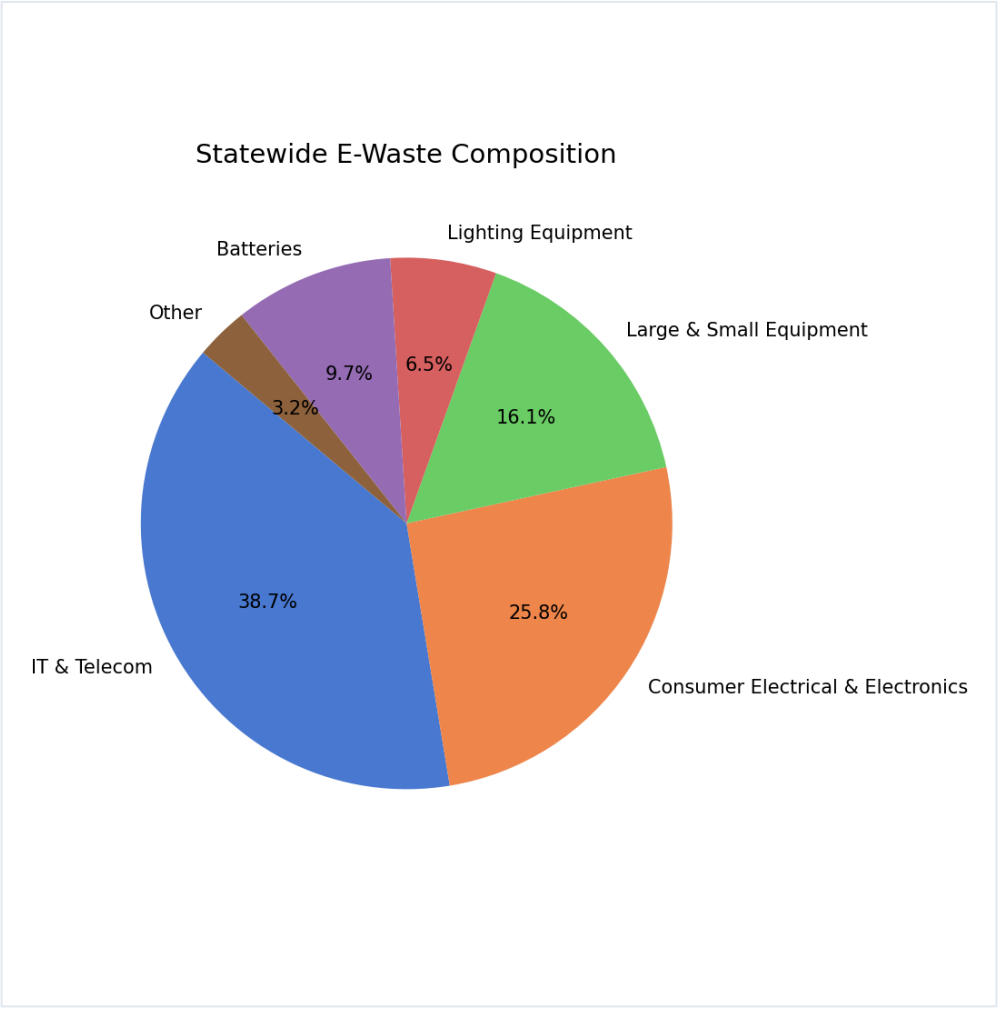
District	Generated (kg)	Collected (kg)	Collection Rate	Registered Centers
STATE TOTAL	2408345	184824	7.67%	15.0
Kangra	468124	75504	16.13%	3.0
Shimla	378514	40700	10.75%	3.0
Mandi	309931	9998	3.23%	1.0
Solan	269849	29016	10.75%	3.0
Sirmaur	246383	5299	2.15%	1.0
Una	161564	5212	3.23%	1.0
Chamba	160915	5191	3.23%	1.0
Hamirpur	140977	4548	3.23%	1.0

District	Generated (kg)	Collected (kg)	Collection Rate	Registered Centers
Kullu	135749	4379	3.23%	1.0
Bilaspur	118407	3820	3.23%	0.0
Kinnaur	13039	841	6.45%	0.0
Lahaul & Spiti	4893	316	6.46%	0.0

> **Key Insight:** There is a stark mismatch in the top-generating districts. For instance, the highest-generating districts are producing vast quantities of e-waste, yet their actual collection rates hover well below 20%. The lack of registered collection centers in rural and semi-urban districts further exacerbates this issue.

4. E-Waste Composition Profile

Understanding the composition of the waste stream is crucial for targeting specific recycling technologies and EPR obligations.



Note: Composition is heavily skewed towards IT & Telecom equipment and Consumer Electricals, which represent the bulk of urban electronic consumption. These items possess high material recovery value (gold, copper, aluminum) but also pose significant heavy-metal toxicity risks if mishandled by the informal sector.

5. Strategic Gap Analysis & Recommendations

A. Collection Infrastructure Deficit

Observation: With a state-wide collection score of just 2.9 out of 20, the physical infrastructure for take-back systems is critically deficient.

- **Recommendation 1:** Mandate Local Urban Bodies (ULBs) to establish at least one permanent e-waste drop-off bin in every municipal ward.
- **Recommendation 2:** Partner with major electronic retailers to enforce mandatory take-back counters (point-of-sale collection) under the EPR framework.

B. Complete Absence of Informal Sector Integration

Observation: The tracker recorded a score of 0 for informal sector integration. The majority of collection is currently being handled by unorganized scrap dealers who dismantle electronics using hazardous methods (acid baths, open burning).

- **Recommendation 1:** Launch a state-funded skill-development program to train informal aggregators in safe dismantling techniques.
- **Recommendation 2:** Create a formal registration portal for 'Kabadiwalas', offering them financial incentives or protective gear to channel collected e-waste directly to the authorized recyclers in industrial zones rather than processing it themselves.

C. Data Transparency and Inventory Updation

Observation: Inventory baseline data (Score: 0/15) is outdated or incomplete. Without accurate district-wise baselines, setting EPR targets is ineffective.

- **Recommendation 1:** Implement a mandatory digital reporting ledger for all bulk consumers (IT parks, government offices, hospitals) to file annual e-waste disposal returns.
- **Recommendation 2:** Conduct a fresh state-wide survey to update population-based generation multipliers specifically for rural vs. urban sectors.
